

Siva Prasad Geetha Nagarajan

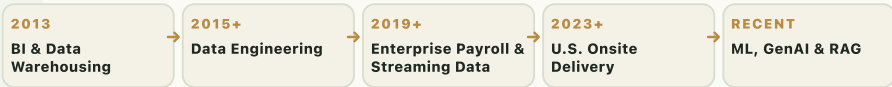
Senior Data & ML/AI Engineer

Data Engineering • Real-Time Streaming • Machine Learning • Generative AI • RAG • Business Intelligence

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PROFILE

Senior Data and ML/AI Engineer with 13+ years of experience designing enterprise data platforms, real-time streaming pipelines, analytical solutions, machine learning applications and Generative AI systems. Brings hands-on U.S. onsite client-delivery experience, working directly with business, tax, product and engineering stakeholders to discover requirements, build rapid POCs, demonstrate solutions and refine them using real-world feedback. Combines deep engineering expertise in Python, SQL, Kafka, cloud data services and BI with applied experience using Gemini, Vertex AI, AWS Bedrock, embeddings and RAG across financial, payroll and compliance workflows. Experienced in progressing ambiguous business problems from discovery through validation, human-in-the-loop design and production readiness.



13+

Years across BI, data, ML
and AI

200+

Streaming and data-
delivery pipelines

\$3.4M

Potential residual
anomalies identified
annually

7→3

Analysts required for the
review workflow

PROFESSIONAL EXPERIENCE

Data & ML/AI Engineer

Apr 2023 — Present

Sutherland Global • formerly ATMECS Global • Phoenix, AZ • Client: Entertainment Partners

Joined ATMECS Global in April 2023 and continued the same client engagement following the Sutherland acquisition.

- Lead discovery sessions with tax, business, product and engineering stakeholders to clarify ambiguous requirements, evaluate feasibility, demonstrate working solutions and refine scope based on real-world data behavior.
- Design and build enterprise data and ML/AI solutions across production, pilot and proof-of-concept initiatives using Python, Gemini, Vertex AI, AWS Bedrock and cloud data services.
- Piloted an AI-assisted Tax Incentives platform across 47 Georgia productions, combining General Ledger, payroll, employee and production data to identify approximately \$500K–\$1M in potentially non-qualified costs per production for analyst validation before filing.
- Shifted tax-incentive review from post-production analysis to daily exception monitoring, reducing manual-review staffing requirements from seven analysts to three while retaining human validation for high-risk cases.
- Deployed a production anomaly-detection system for film and television residual payments, identifying approximately \$3.4M in potentially anomalous transactions annually for investigation and remediation.
- Designed and demonstrated an AWS Textract-based invoice-processing POC using OCR confidence, validation rules and invoice thresholds to recommend straight-through approval or manual review; transferred the architecture and findings to a specialized productionization team.

Python • SQL • Gemini • Vertex AI • AWS Bedrock • AWS Textract • PostgreSQL • AWS EC2 • AWS MSK • Qlik Sense

Associate Technical Lead

Dec 2015 — Mar 2023

ATMECS Inc. • Hyderabad, India • Clients: Entertainment Partners, Citrix & others

- Led an eight-member Data Engineering and streaming team across architecture, hiring, mentoring, delivery, production support and stakeholder communication.
- Led the architecture and production operation of more than 200 streaming and data-delivery pipelines supporting payroll distribution to ERP, analytics and reconciliation systems.
- Implemented event-driven and change-data patterns using Apache Kafka, AWS MSK, AWS Kinesis and MongoDB Change Streams.
- Built Qlik Sense applications for operational and leadership decision-making, including real-time visibility into pipeline health, throughput and failures.
- Built workforce analytics dashboards that helped studios compare crew availability, role experience and compensation benchmarks for production staffing decisions.
- Designed a data warehouse built on Google BigQuery for Citrix and collaborated with Entertainment Partners R&D teams on machine learning and data-science pipelines.
- Supported production through monitoring, defect resolution, reconciliation, data validation and customer demonstrations.

Python • SQL • Kafka • AWS MSK • AWS Kinesis • MongoDB • DocumentDB • AWS Glue • AWS EC2 • BigQuery • PostgreSQL • Qlik Sense • Sqoop • Pig • Oozie

BI Developer / Software Engineer

Jul 2013 — Oct 2015

Virtusa Corporation · Hyderabad, India · Clients: Network Rail, IMS & Citi

- Designed OBIEE repository layers, semantic models, dashboards, interactive reports and complex ad hoc queries for Network Rail and IMS.
- Built unit-cost modeling reports and dashboards supporting infrastructure cost analysis for Network Rail.
- Delivered SAP BusinessObjects and Tableau reporting solutions for Citi.
- Implemented object- and data-level security, Active Directory authentication, user administration, performance tuning and cache management.
- Integrated OBIEE reports with Oracle APEX and produced data-lineage, technical and implementation documentation.

OBIEE 11g · SAP BusinessObjects · Tableau · Oracle 11g · Oracle APEX · Informatica · JavaScript · Linux · SQL

SELECTED PROJECTS

Tax Incentives Intelligence

Pilot · 47 Georgia productions · Human-in-the-loop review

- Ran stakeholder discovery to combine General Ledger, payroll, employee and production data, surfacing potentially misclassified transactions before filing.
- Applied deterministic tax rules, Gemini on Vertex AI, embeddings, vector retrieval and RAG to provide supporting context for analyst decisions.
- Designed daily exception workflows so high-risk cases remained reviewable and traceable rather than relying on unrestricted LLM output.

Residual Payment Anomaly Detection

Production · Payment integrity

- Combined statistical and machine-learning signals with domain rules to detect unusual payment patterns across film and television residuals.
- Prioritized high-risk exceptions for operational investigation while controlling manual-review volume.
- Integrated detection outputs into a human-review workflow for remediation and feedback.

Invoice Automation with AWS Textract

POC · OCR extraction and approval recommendations

- Extracted invoice fields with AWS Textract and evaluated confidence, validation rules and invoice thresholds.
- Recommended straight-through approval for low-risk invoices and manual review for uncertain or higher-risk cases.
- Demonstrated the POC to stakeholders and documented feasibility findings and transition considerations for handwritten and non-standard vendor formats.

Enterprise Payroll Data Streaming

200+ pipelines · Multi-consumer delivery

- Supported payroll distribution to ERP, analytics and reconciliation systems through streaming and data-delivery pipelines.
- Used Kafka, AWS MSK, AWS Kinesis and MongoDB Change Streams for reliable event and change-data movement.
- Built Qlik Sense operational dashboards for pipeline health, throughput, failures and proactive issue resolution.

Workforce Analytics for Production Staffing

Dashboards for crew availability & pay benchmarking

- Built analytics dashboards helping studios identify available, qualified crew for open production roles.
- Modeled pay benchmarks by role and experience level to support fair, competitive compensation decisions.
- Presented crew availability, role experience and compensation benchmarks to support human-led production staffing decisions.

BI & Decision Intelligence

From semantic models to executive insights

- Built enterprise BI solutions using Qlik Sense, OBIEE and Tableau across operational and leadership use cases.
- Created data models, reports, dashboards and analytical experiences for technical and non-technical audiences.
- Connected traditional BI foundations with modern cloud data, streaming and AI-enabled workflows.

TECHNICAL CAPABILITIES

Professional & Production Experience

Technologies used across enterprise delivery, production systems, client projects or formal pilots.

Data Engineering & Streaming

Python SQL ETL / ELT Data Modeling Data Quality Apache Kafka AWS MSK AWS Kinesis MongoDB Change Streams FastAPI Pydantic Alembic pandas

AI / ML / GenAI

Machine Learning Gemini Claude Vertex AI AWS Bedrock AWS Textract Embeddings RAG Vector Search pgvector Prompt Design LLM Evaluation scikit-learn XGBoost LightGBM RapidFuzz PyTorch TensorFlow MLflow Optuna

Cloud, Databases & BI

AWS GCP EC2 S3 Glue Lambda BigQuery PostgreSQL MongoDB DocumentDB Oracle SQL Server Redshift DB2 MySQL Apache Solr Qlik Sense OBIEE Tableau SAP BusinessObjects Docker Kubernetes

Hands-On Prototyping & Development

Used in technical exploration, POCs, learning projects or solution experiments.

Hugging Face LangChain LlamaIndex
FAISS PySpark SageMaker Airflow
Tool Calling Agentic Workflows SFT
LoRA / QLoRA

LEADERSHIP & DELIVERY

- Led an eight-member Data Engineering and streaming team.
- Owned architecture, technical reviews, mentoring and production troubleshooting.
- Worked directly with U.S. client stakeholders to convert business needs into POCs and scalable solutions.
- Explained AI limitations, review requirements, risks and production-readiness trade-offs to technical and business audiences.
- Delivered using Agile practices across sprint planning, stand-ups and iterative releases.

PROFESSIONAL DEVELOPMENT

Production AI Systems

Transformer foundations, LLM evaluation, agentic workflows, tool calling, model adaptation, human-in-the-loop design, monitoring and AI solution economics.

Open-Weight Model Evaluation

Evaluating open-weight and proprietary LLMs for reasoning quality, tool use, deployment flexibility, privacy, latency and cost trade-offs.

Client-Facing AI Solution Delivery

Expanding client-facing solution capabilities through focused study and practice in stakeholder discovery, requirements engineering, rapid POC development, technical demonstrations and enterprise RAG architecture — building toward Forward Deployed Engineer and client-embedded AI solution roles.

EDUCATION

M.Tech in Data Science & Engineering

BITS Pilani · Work Integrated Learning Programme · 2019–2022 · Completed while working full-time

B.E. in Electrical & Electronics Engineering

Anna University, Chennai